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A Circular Approach for Multiscale Changepoint Detection in Nonstationary Climate Data

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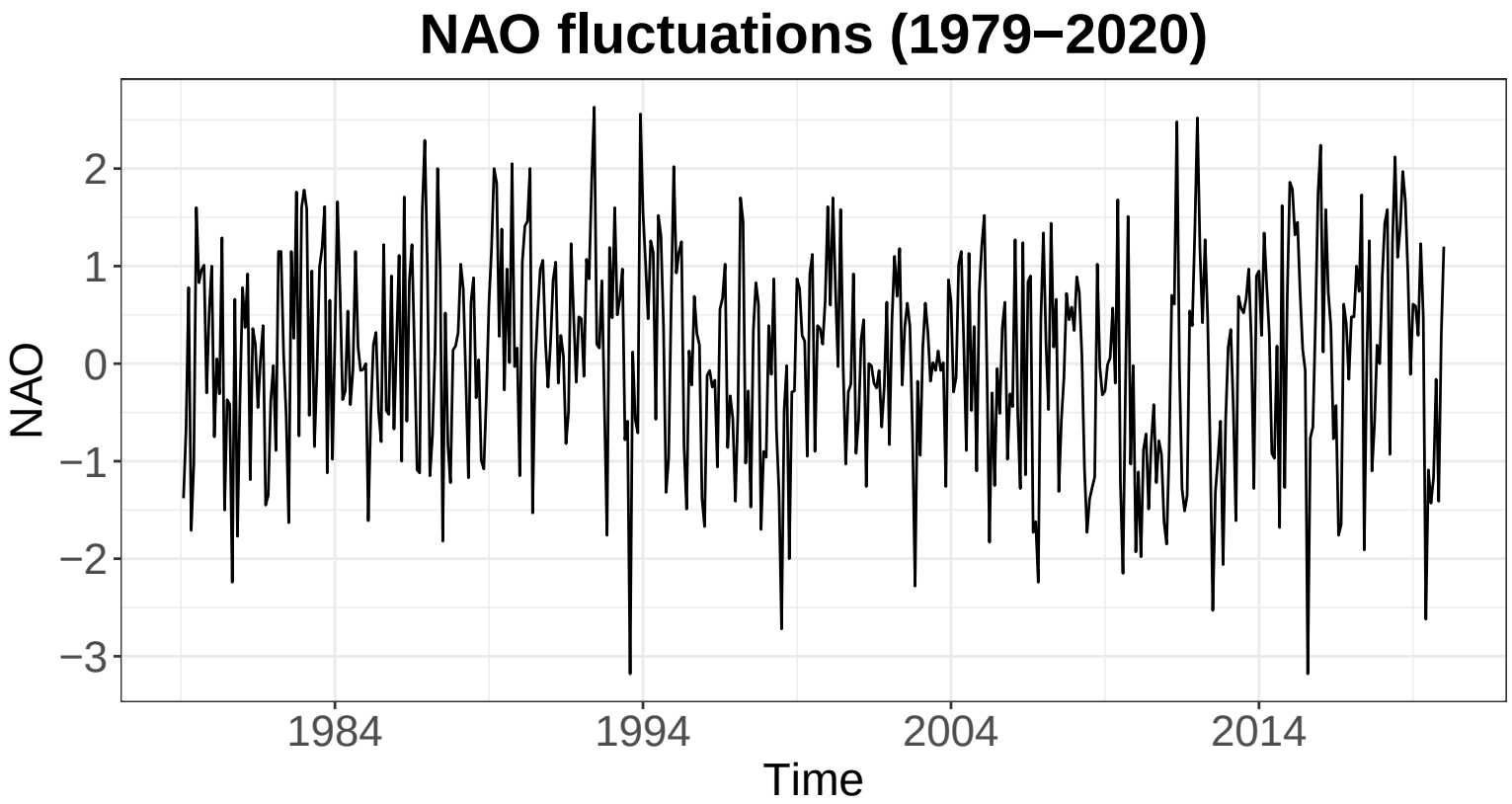
Outline

- Motivation
- Micro changepoint model
- RJMCMC Implementation
- Macro changepoint model
- Simulations
- Application

Motivation

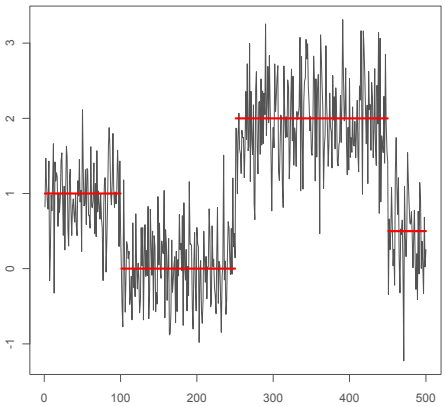
NAO

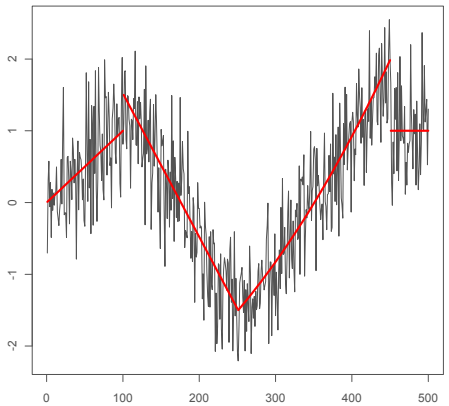
- How can we make the most of highly granular time series?
- How can we study changes in periodic and cyclical behaviours?

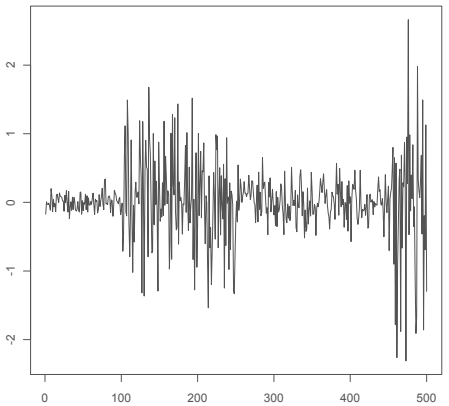


What are changepoints?

For data $y_1, \ldots, y_n$, a changepoint is a location τ where the statistical properties of $y_1, \ldots, y_\tau$ are different from $y_{\tau+1}, \ldots, y_n$ in some way. Traditional changepoints include: mean, regression, variance.







Existing Changepoint Methods

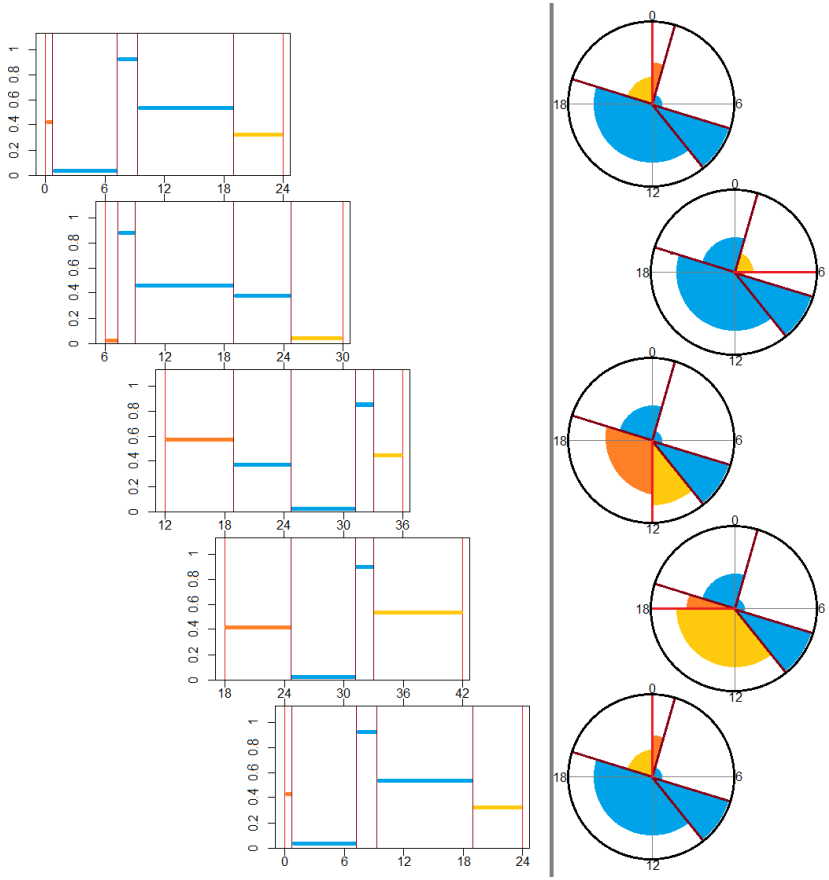
- Binary segmentation & PELT
  - Assumes conditional independence ...
  - ... but dependence exists across days.

- Multivariate time series
  - Stack days to align changepoints ...
  - ... but what should 'time zero' be.

Existing Changepoint Methods

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- Circular changepoint
  - For circular data (e.g. direction) ...
  - ... but time axis is linear.
- Circular Binary Segmentation
  - local time axis is circular ...
  - ... but just joins up ends not each day.
- Cyclic-HMM – Achieves periodicity via a structured transition matrix ... but period duration is stochastic.



Our Contribution

Create method for within-period changepoint detection

- This talk for real-valued data - original RSSC paper is binary data
- Bayesian method with truncated poisson prior for cpt locations
- RJMCMC for fitting

Extend the within-period to across period changes too.
Create layered visualizations for ease of inference.

We denote changepoints as micro (within) and macro (across) level changepoints.

Micro level changepoint detection

Model

- $\{X_t\}_{t \geq 0}$ — Real-valued time series.
- $N \in \mathbb{N}$ — Given period length.
  - $t = cN + i$ where $c \in \mathbb{N}_0$ and $i \in \{0, 1, \ldots, N-1\}$.
- $m \geq 1$ — Number of segments partitioning the period.
- $\tau_1, \ldots, \tau_m \in \{0, 1, \ldots, N-1\}$ — Changepoint positions.
  - Ordered: $\tau_{j-1} < \tau_j$.
  - Periodic: $\tau_{m+1} = \tau_1$.
  - Minimum separation: $\tau_j - \tau_{j-1} \geq l$ for $l > 0$.
- μ_j for $j \in \{1, \ldots, m\}$ — Segment means.
- σ_j^2 for $j \in \{1, \ldots, m\}$ — Segment variances.

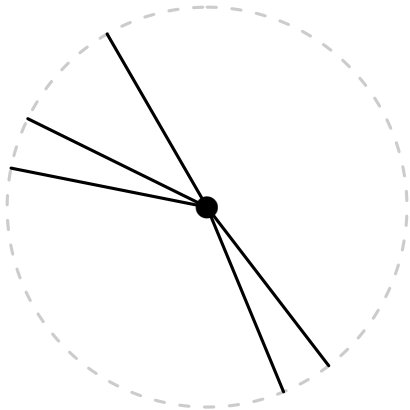
$X_t \sim \text{Normal}\left(\mu(t), \sigma^2(t)\right)$ for $t \geq 0$ where
$\mu(t) = \sum_{j=1}^{m} \mu_j \mathbb{I}\{i \in (\tau_{j-1}, \tau_j]\}$ and $\sigma^2(t) = \sum_{j=1}^{m} \sigma_j^2 \mathbb{I}\{i \in (\tau_{j-1}, \tau_j]\}$.

Prior

- $m \sim \text{Truncated} - \text{Poisson}\left(1,\ \{1, \ldots, \lfloor \frac{N}{l} \rfloor\}\right)$.
- $\mu_j | \sigma_j^2, \boldsymbol{\tau}, m \sim \text{Normal}(d, c\sigma_j^2)$ for $j = 1, \ldots, m$.
- $\sigma_j^2 | \boldsymbol{\tau}, m \sim \text{InvGamma}(a, b)$ for $j = 1, \ldots, m$.
- Transform $\boldsymbol{\tau}$ to excess segment length:
  - $\delta_j = \tau_j - \tau_{j-1} - l$ for $j = 1, \ldots, m$.
  - $(\delta_1, \ldots, \delta_m) \sim \text{Dirichlet} - \text{Multinomial}(N - lm,\ \alpha \mathbf{1}_m)$
- $\tau_1 \sim \text{Uniform}\{0, \ldots, N-1\}$.

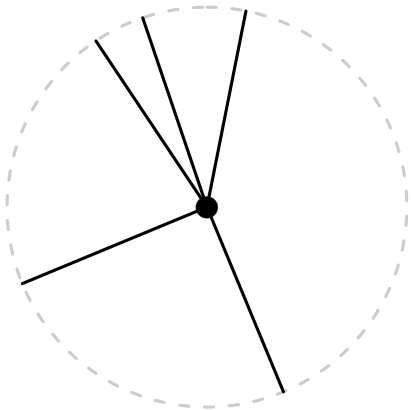
Cluster

$\alpha = 0.5$



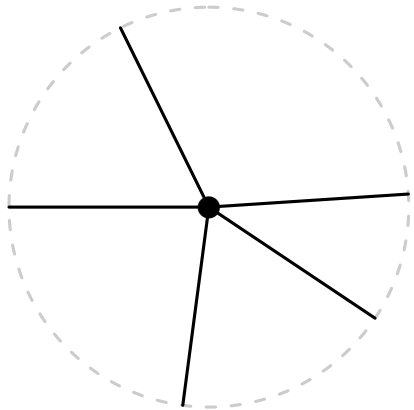
Random

$\alpha = 1$



Repel

$\alpha = 2$



Posterior

Segment probabilities: Posteriors conditional on a sampled number and location of changepoints are easy to compute via conjugacy.

Number of segments and changepoint positions:

$$\pi(\boldsymbol{\tau}, m|\boldsymbol{x}) \propto \begin{cases} B(1+\sum_t x_t,\ 1+\sum_t(1-x_t)) & m=1, \\ \prod_{j=1}^{m}\left[\frac{B(1+s_j,\ 1+n_j-s_j)}{(\alpha+\delta_j)B(\alpha,\ \delta_j+1)}\right]\frac{(m\alpha+N-lm)B(m\alpha,\ N-lm+1)}{m!N} & \text{o/w}, \end{cases}$$

where $B(\cdot,\cdot)$ is the beta function.

- Trivial to evaluate $\pi(\boldsymbol{\tau}, m|\boldsymbol{x})$ for a given changepoint vector.
- Support size is potentially impractically large:
  - For $N = 96$ and $l = 4$, then there are $\sim 27.3 \times 10^{12}$ events.

Reversible Jump Monte Carlo Markov Chain (RJMCMC)

RJMCMC

- Permits exploration of varying dimensional spaces.
- Need to satisfy 'dimension-matching' condition.
- Focus inference on the marginal posterior $\pi(\boldsymbol{\tau}, m|\boldsymbol{x})$.

Algorithm

1. Sample a segment $j \in \{1, \ldots, m\}$ uniformally.
2. Define the proposal set $\mathcal{P}_j(\boldsymbol{\tau}, m)$.
3. Sample a changepoint vector from the proposal pmf:

$$q(\boldsymbol{\tau}', m'|\boldsymbol{\tau}, m) \propto \pi(\boldsymbol{\tau}', m'|\boldsymbol{x}) \mathbb{I}\left\{(\boldsymbol{\tau}', m') \in \mathcal{P}_j(\boldsymbol{\tau}, m)\right\}.$$

4. Accept/Reject proposal. Return to 1.

- Evaluate $\pi(\boldsymbol{\phi}|\boldsymbol{\tau}, m, \boldsymbol{x})$ given the changepoint posterior samples.

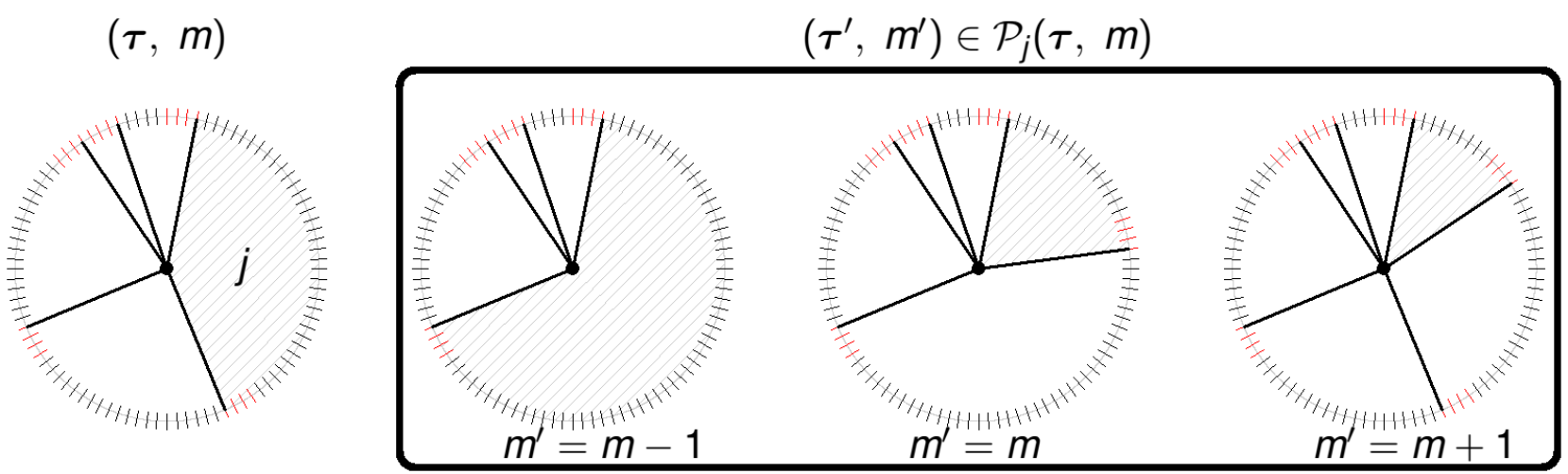
Proposal Set

Proposal set, $\mathcal{P}_j(\boldsymbol{\tau}, m)$, contains three move types:

- Combine sampled segment with the next.
- Contract or extend the length of the current segment.
- Partition current segment into two.

Unique mapping

For each $(\boldsymbol{\tau}', m') \in \mathcal{P}_j(\boldsymbol{\tau}, m)$ excluding $(\boldsymbol{\tau}, m)$, there exists a unique segment j' such that $(\boldsymbol{\tau}, m) \in \mathcal{P}_{j'}(\boldsymbol{\tau}', m')$.



Macro Level Changepoint Detection

Macro Level Changepoints

- Treat the micro-level detection as a "fit" for a segment ...
- ... thus we can use traditional, linear time algorithms.

We use PELT which optimizes:

$$\sum_{i=1}^{q+1}[\mathcal{C}(y_{(\alpha_{i-1}+1):\alpha_i})] + \beta q. \tag{1}$$

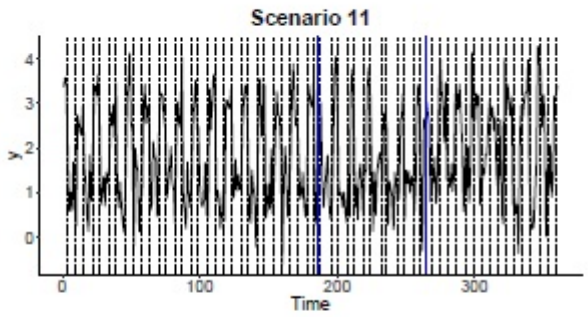
where q is the number of macro level changepoints and $\mathcal{C}(\cdot)$ is the MAP likelihood:

$$\mathcal{C}(y_{(\alpha_{i-1}+1):\alpha_i}) = -2\sum_{j=1}^{\hat{m}_i}\sum_{t\in B^*_{i,j}} \log f(y_t|\hat{\mu}_{i,j}, \hat{\sigma}^2_{i,j})$$

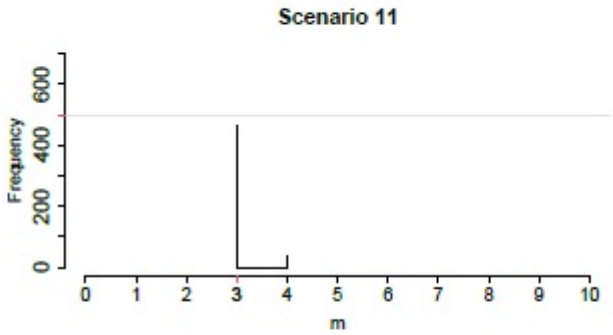
Simulations

See pre-print for full simulation study.

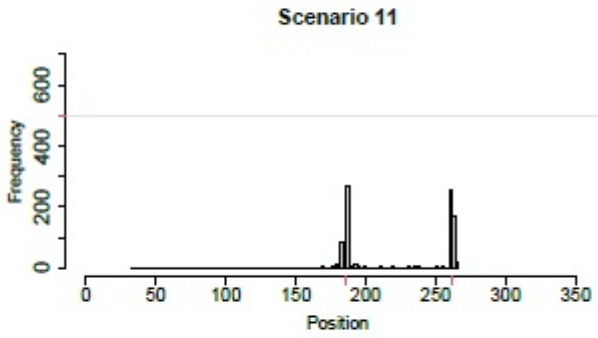
Simulation Results



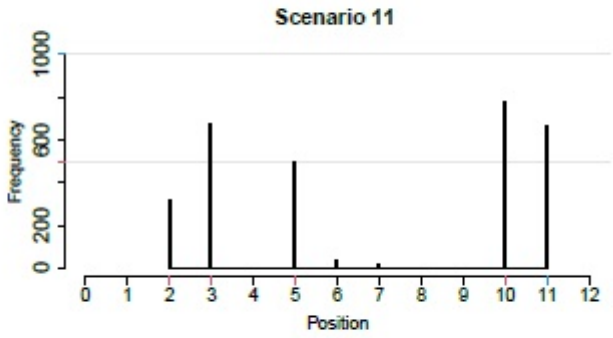
(a) Realisation



(b) Number of Segments



(c) Macro-changes Positions



(d) Micro-changes Positions

Figure: Simulation results for scenario 11: 2 macro-change points at 6th month of 15th period and 10th month of 21st period, micro-changes at months 3,10, 2,11 and 5,11 for the segments 1,2 and 3 respectively.

Application

Application Results:NAO

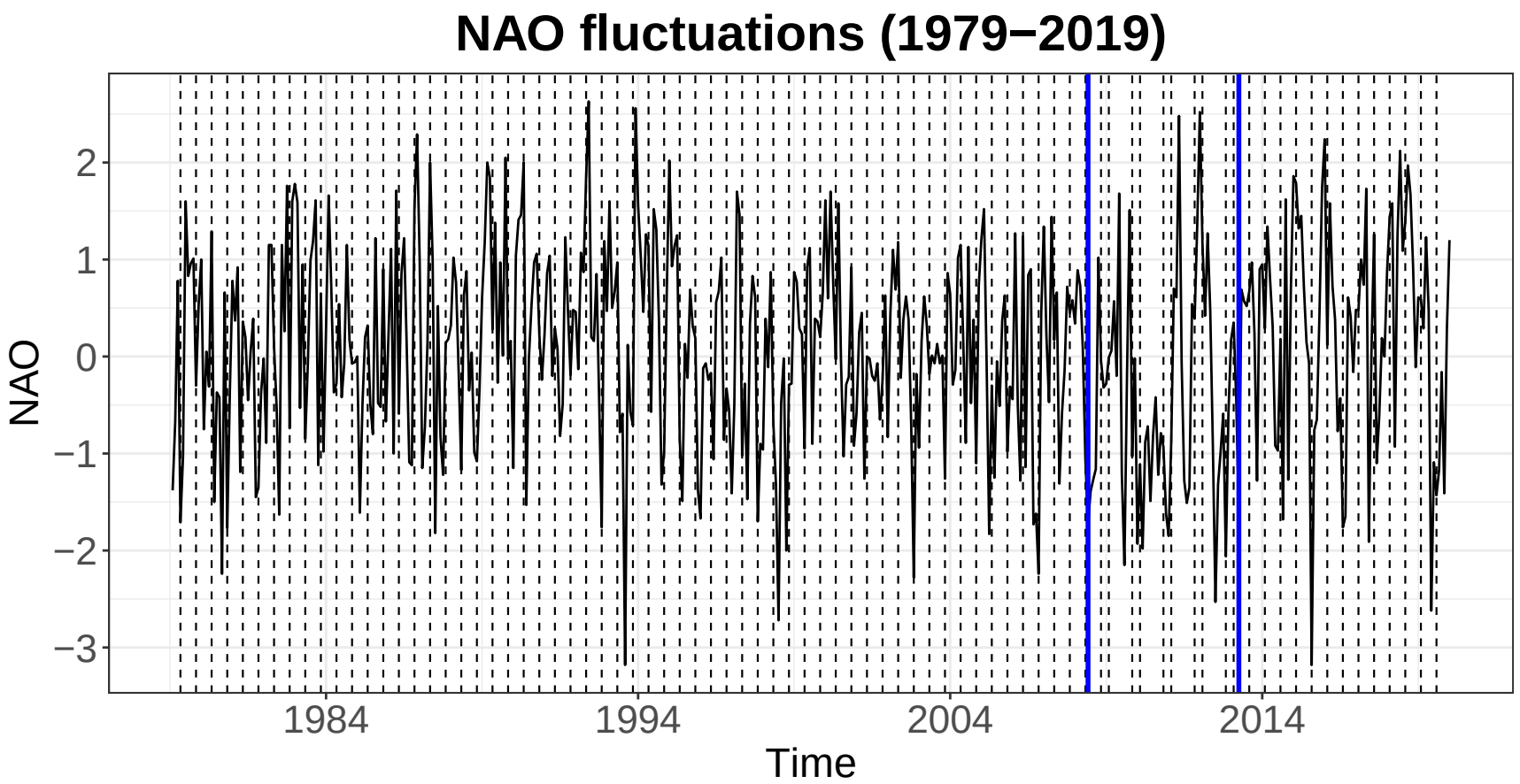


Figure: NAO data and Changepoint positions using $pen = 4\log(n)$

Application Results:NAO

Macro-level segment	Micro-level changepoints
1979-2008	April and October
2008-2013	May and August
2013-2019	April and October

Table: Macro- and micro-level changepoint positions for the NAO index.

Discussion

Overview:

- A new approach for identifying within-period changepoint events for time series.

  Taylor, S. A. C., Killick, R., Burr, J. and Rogerson, L. (2021)

  'Changepoint Detection within Periodic Binary Time Series'. JRSS:Series C.

- Pooling of evidence across multiple periods by applying a circular perspective to time.

- Periodic changepoint patterns of real-valued time series.
- Introduced the micro-macro level changepoint approach

  Ushakova, A., Taylor, S. A. C., Killick, R. (2021)

  'Multi-level Changepoint Inference for Periodic Data Sequences'. Under Review.